Where do my cells sit?

A habit worth having, and the loop that pays for it.

← all documents · Kat Titterton · San Francisco

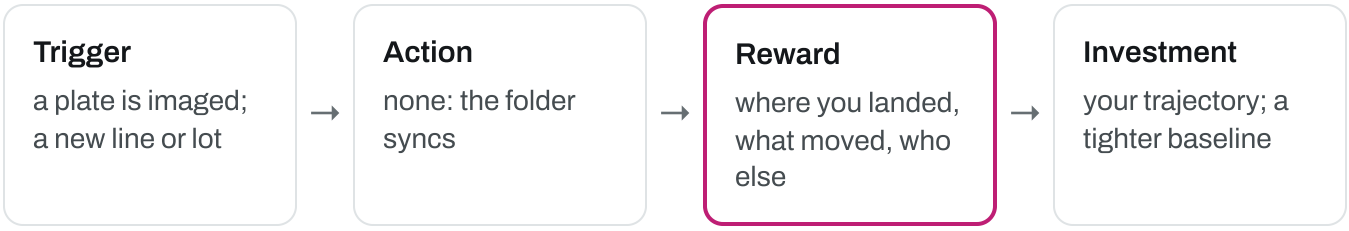
Vision

Every lab running a cell line should know, every time it images a plate, where its cells sit among everyone else's, and what to do if they have moved. The model is Strava's segments: your run placed among everyone who ran the same stretch of road. Here the road is the cell line, and the field is every lab that ran it.

The product earns its habit the way a bathroom scale or a credit score does: the number is about you, it changes, and nobody knows it until they look. It never earns it the way a feed does. Everything below serves one rule: **uploading must be effortless, and the answer must be worth looking at.**

01 The habit loop

Fig. 1 **Every upload makes the next answer worth opening**



↺ back to Trigger: the reference grows, the next reading is sharper

STEP	IN DATUM LINE	DESIGN RULE
Trigger	the microscope finishes a plate; a new line or lot arrives	the trigger is the instrument, not an email
Action	nothing: the export folder syncs	any click is a leak; the reference starves on friction
Reward	where you landed, which features moved, who else moved	real uncertainty: nobody knows their position until they look
Investment	your history becomes a trajectory; your baseline tightens	leaving costs your own past, which is honest switching cost

02 Six magic moments

Each moment turns an image the lab already has into something it could not see before, and each is ordered by when a new customer meets it.

#	MOMENT	WHAT THE LAB SEES	WHEN
1	**Your past, diagnosed**	old plates dragged in on day one appear as a path across the field map: "you left the envelope in March"	first ten minutes
2	**Show the cells, not the score**	when a plate moves, its nuclei next to their nearest neighbours from labs in distribution: same size, more texture	every flag
3	**The fix**	what the in-distribution labs do differently: seeding density, media or serum lot	after a flag
4	**A forecast**	"at your current drift, this line leaves the envelope in about five weeks"	after three readings
5	**Someone else's plate saves your screen**	a lot you use is drifting at other labs, before your own plates show it	once labs share lots

#	MOMENT	WHAT THE LAB SEES	WHEN
6	**The certificate as currency**	a verifiable link for a methods section, a CRO handoff, a grant report	when results leave the lab

Moments 1 to 4 work for a single lab on day one, against the public reference. Moments 5 and 6 exist only once the network does, and no competitor can copy them without the labs.

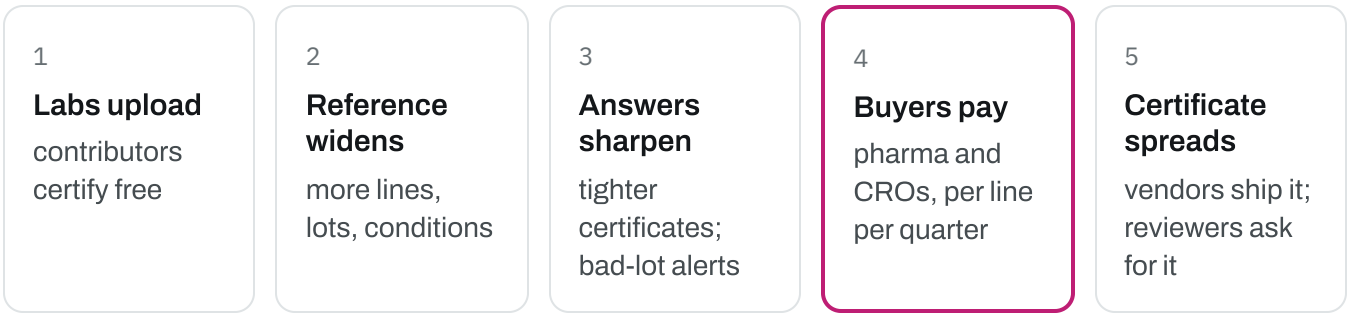
03 The revenue loop

The real product is the standard: one map every lab's cells are placed on, and the benchmark every method is graded against. It has three layers. **Community**, open: the map, the box, the code, the public reference, labs by number, streaks and ranks; it stays fully usable for free, forever, or the standard dies. **Data access**, paid: the same data structured, comparable, pre-cleaned on request, with provenance and the certificate attached; the API, not the website. **The benchmark**, co-built: industry partners contribute failures and get their methods graded, segmentation to bioactivity, so the labs that make data and the companies that buy it grade against one set.

A use case the map was not built for: integrity. Cells cannot lie about where they came from. The classifier that names the lab is also a provenance check on any plate that claims an origin; the known-answer check is an integrity check on any plate map; and an untreated well is a canary that moves when something in the room changes. Control wells only, so nothing about agents, compounds or targets leaves a lab.

People and agents, one map. People get the box, the glimpse of everyone else's cells and the game. Agents get the API: structured, comparable data with provenance, a coordinate system to reason in, and a benchmark to be graded on. Every drop sharpens the map people read and agents train on; every graded method raises the benchmark both are held to. Everyone learns, and the map grows with them.

Fig. 2 **Free uploads build the reference that paid certificates sell**



↺ 5 brings more labs to 1; revenue funds bench time and people

The habit loop feeds the revenue loop. Free contributors widen the reference; a wider reference sharpens every certificate and makes bad-lot alerts possible; sharper answers are what pharma and CROs pay for; paid and shared certificates spread through vendors and reviewers, which brings more labs in.

BUYER	PAYS FOR	PROPOSED PRICE	WHY THEY KEEP PAYING
Academic labs and cores	nothing, if they contribute	free for contributors, \~$500–1,000 per line-quarter otherwise	their own trajectory and the diagnosis of their past
Pharma and CRO screening	certificates, forecasts, the fix	\~$5–15k per line-quarter, or a site license	one late-caught line cost \~$2.6M; a quarterly reading is insurance
Cell banks and vendors	a lot certificate to ship with each lot	per lot	customers start asking for it
Everyone, later	bad-lot alerts	subscription	only the network can produce them

The quarter is a starting cadence: in the one lab measured, a baseline is stale within weeks. Pricing is proposed, not validated; the roadmap doc carries the anchors.

Who pays, exactly. The buyer of cell data it did not make: an AI-bio company or a pharma paying a CRO for a screen. The CRO uploads for free because its customer asks for the certificate; the customer pays per certificate on the data it bought. Labs that only ever upload still get the map, the certificate and the game, and every one of them makes the map denser for the paying side.

04 A healthy habit, not a trap

The product dies the day a lab hides its worst plate, because the worst plates teach the field most. Every design choice protects honest uploads.

DO	DON'T	WHY
Show each lab its position privately; publish only the field	public leaderboards of labs	shame makes labs withhold bad plates, which poisons the reference
Give credit for uploading flagged plates	penalise being out of distribution	failures are the most valuable data
Keep a record of certified quarters	"don't break your streak" nudges	a streak rewards hiding the plate that would break it
Explain every flag with cells and a suggested fix	a bare red dot	a verdict without a reason feels like judgement, not help
Let the instrument trigger uploads	notifications to come back and check	the reward is the answer, not the visit

05 The end state: from loved to required

The strongest habit is one an institution asks for. Identity testing got there when journals and funders began asking labs to show their cells were what they claimed. A behaviour certificate can follow the same path: first reviewers ask for it, then CRO contracts include it, then a vendor lot ships with it by default.

That lock-in has a cost: a required certificate reads as compliance, and compliance is rarely loved. So the order matters. Build the loved product first (the map, the cells, the fix, the warning) so that when it becomes required, labs already want it.

06 Metrics

North star: the share of a customer's plates that upload with no human action. If it is high, the loop turns; if people have to remember, nothing else matters.

METRIC	WHAT IT TELLS YOU	LOOP
Automatic upload share	the habit exists without effort	habit
Flagged plates uploaded, not deleted	labs trust the product with bad news	honesty
Time from upload to first look	the answer is worth opening	reward
Lines per lab, labs per line	the reference is widening	network
Labs sharing a lot	bad-lot alerts are possible	network
Paid line-quarters, renewal rate	the certificate is worth money, again	revenue
Certificates cited outside the lab	the certificate has become currency	spread

The game

Every batch a lab sends is ranked against every batch the field has seen, on its untreated wells alone: how typical the cells are, how alike the plates are, how steady the lab is from batch to batch, and cells per well against the field. Nothing in the score touches a treated well or a result, so the only way to climb is better culture and imaging practice. A lab sees its rank, its personal best, its streak of batches in distribution, the one measure that would move it up most, and the lab to beat. On the public data one lab holds 9 of the top ten and a streak of 33 (results/e14). Segments are per cell line; a quarter is a season; badges are for the centre, the tightest plates and the longest streak. Names on the board are the lab's choice: a lab can compete in public or watch in private.